

$$b) \left[\begin{array}{cc|ccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 5 & 12 & 3 & -5 & 6 & 5 \end{array} \right] \xrightarrow[5 \times \text{Row } 1]{\text{Row } 2 = \text{Row } 2 -} \left[\begin{array}{cc|ccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 2 & 8 & -10 & -4 & -10 \end{array} \right]$$

$$\xrightarrow[\times 1/2]{\text{Row } 2 = \text{Row } 2} \left[\begin{array}{cc|ccc} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{array} \right] \xrightarrow[-2 \times \text{Row } 2]{\text{Row } 1 = \text{Row } 1} \left[\begin{array}{cc|ccc} 1 & 0 & -9 & 11 & 6 & 13 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{array} \right]$$

c) Use matrix algebra to show that if A is invertible and D satisfies $AD = I$, then $D = A^{-1}$.

Solution: By definition of an invertible matrix, there exists a matrix A^{-1} such that $A^{-1}A = AA^{-1} = I$.
 $AD = I \Rightarrow A^{-1}(AD) = A^{-1}I \Rightarrow (A^{-1}A) \cdot D = A^{-1}I$ (Theorem 2(a))
 $\Rightarrow I \cdot D = A^{-1}I$ (By defn.) $\Rightarrow D = A^{-1}$ (Theorem 2(e))

In Exercises 9 and 10, mark each statement True or False. Justify each answer.

9a) In order for a matrix B to be the inverse of A , both equations $AB = I$ and $BA = I$ must be true.

Solution: This statement is True by the defn. of invertible matrix on Page 105.

b) If A and B are $n \times n$ and invertible, then $A^{-1}B^{-1}$ is the inverse of AB .

Solution: By Theorem 6(b) on Page 107, we know that $(AB)^{-1} = B^{-1}A^{-1}$.
 On Page 100, just above example 7, ~~and~~ it is mentioned that matrix multiplication do not necessarily commute. $\therefore (AB)^{-1} \neq A^{-1}B^{-1}$.
 $\therefore$ The statement is False.

c) If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad - bc \neq 0$, then A is invertible.

Solution: By Theorem 4 on Page 105, the statement is False.

d) If A is an invertible $n \times n$ matrix, then the equation $Ax = b$ is consistent for each b in $\mathbb{R}^n$.

Solution: By Theorem 5 on Page 106, and by defn. of consistency on Page 4, the statement is True.